

Summary and Recommendations

Objective

The objective of this analysis is to identify the key drivers of customer churn and provide data-driven recommendations to improve retention. This was achieved through **exploratory data analysis (EDA)** and the creation of multiple **interactive and comparative visualizations** to uncover patterns and trends in customer behavior.

Overview of Dataset

The dataset includes customer demographics, service subscriptions, tenure, contract types, and payment methods. The overall churn rate is approximately **26.5%**, meaning that about **1 in every 4 customers** leaves the service.

Visualization Approach

A variety of **charts and plots** were created to effectively analyze and communicate insights:

- **Count Plots** to compare churn distribution across categorical variables (e.g., gender, senior citizen, contract type)
- **Stacked Bar Charts (percentage-based)** to show churn proportion relative to total customers
- **Histogram / Distribution Plots** to analyze tenure patterns
- **Multiple Subplots** to visualize the impact of different service features (e.g., Online Security, Tech Support, Streaming services)
- **Grouped Bar Charts** to compare churn behavior across different categories

These visualizations helped in identifying trends quickly and made the analysis more intuitive and actionable.

Key Insights from Visualizations

1. Churn Distribution (Count Plot)

- The churn count plot clearly shows that **26.5% customers have churned**, while **73.5% are retained**.
 - The visual imbalance highlights the need for retention-focused strategies.
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2. Tenure Analysis (Histogram / Bar Chart)

- Tenure distribution charts show that **customers with shorter tenure (0–12 months)** have the highest churn.
 - Customers with **longer tenure (>24 months)** are significantly more stable.
 - The visualization clearly demonstrates that **retention improves over time**.
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3. Contract Type (Stacked Bar Chart)

- A percentage-based stacked bar chart reveals:
 - **Month-to-month contracts** → **~40–45% churn rate**
 - **1-year contracts** → **~10–15% churn**
 - **2-year contracts** → **<10% churn**
 - This chart strongly indicates that **long-term commitment reduces churn**.
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4. Services Impact (Multiple Subplots / Count Plots)

- Subplots comparing services such as:
 - Online Security
 - Tech Support
 - Device Protection
 - Clearly show that customers **without these services have higher churn (~30–40%)**.
 - Customers with these services show significantly **lower churn rates (~15–20%)**.
 - These visuals highlight the importance of **value-added services in retention**.
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5. Internet Service Type (Grouped Bar Chart)

- Charts comparing **DSL vs Fiber Optic** users indicate:
 - **Fiber Optic customers have higher churn (~35–40%)**
 - DSL users show relatively lower churn
 - This suggests potential issues related to **pricing or service quality**.
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6. Payment Method (Count Plot / Stacked Chart)

- Visualizations show:
 - **Electronic check users** → highest churn (~40%+)
 - **Auto-pay users (credit card/bank transfer)** → lower churn (~15–20%)
 - This insight suggests that **payment convenience influences retention**.
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7. Demographics (Count Plot)

- **Senior Citizens** show relatively higher churn compared to non-seniors.
 - **Gender-based charts** show minimal variation, indicating gender is not a strong churn factor.
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Key Findings

- Visual patterns confirm that **early-stage customers are at high risk of churn**
- **Long-term contracts significantly improve retention**
- **Lack of additional services is strongly linked with higher churn**
- **Manual payment methods increase churn probability**
- **Fiber Optic service users require deeper investigation**

Recommendations

- Use insights from visualizations to target **high-risk segments (new customers, monthly contracts)**
- Promote **bundled services** to increase customer engagement
- Encourage adoption of **automatic payment methods**
- Offer incentives for **long-term contract subscriptions**
- Improve experience for **Fiber Optic users**